

ATIVIDADE DE APOIO – ENSINO MÉDIO

Componente Curricular: Física - Vetores

Data: Junho/2025

Professor (a): Rosana

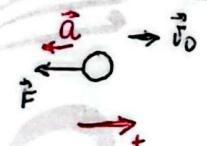
Série: 1ª

Assunto (s): Aplicações da 2ª Lei de Newton - Exercícios - Dinâmica

1. a) $P_T = m \cdot g_T \Rightarrow P_T = 10 \cdot 9,8 \Rightarrow \boxed{P_T = 98 \text{ N}}$
 b) $P_L = m \cdot g_L \Rightarrow P_L = 10 \cdot 16 \Rightarrow \boxed{P_L = 160 \text{ N}}$

2. $P_L = 48 \text{ N} \Rightarrow P_L = m \cdot g_L \Rightarrow 48 = m \cdot 16 \Rightarrow m = \frac{48}{16} \Rightarrow \boxed{m = 3 \text{ kg}}$
 $P_T = m \cdot g_T \Rightarrow P_T = 30 \cdot 10 \Rightarrow \boxed{P_T = 300 \text{ N}}$

3. $m = 5,0 \text{ kg}$
 $v_0 = 24 \text{ m/s}$
 $F = 20 \text{ N}$

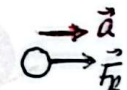


a) $F_R = m \cdot a \Rightarrow F = m \cdot a \Rightarrow 20 = 5 \cdot a \Rightarrow a = \frac{20}{5} \Rightarrow a = 4 \text{ m/s}^2$
 b) $v = v_0 + at \Rightarrow 0 = 24 - 4 \cdot t \Rightarrow 4t = 24 \Rightarrow t = \frac{24}{4} \Rightarrow \boxed{t = 6 \text{ s}}$
 c) $v^2 = v_0^2 + 2a\Delta s \Rightarrow 0 = 24^2 + 2 \cdot (-4) \cdot \Delta s \Rightarrow 8 \cdot \Delta s = 576 \Rightarrow \Delta s = \frac{576}{8}$
 $\Delta s = 72 \text{ m} \therefore d = |\Delta s| \Rightarrow \boxed{d = 72 \text{ m}}$

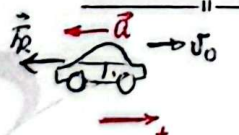
4. $m = 5 \text{ kg}$
 $F_R = 30 \text{ N}$
 $t = 5 \text{ s}$

* $F_R = m \cdot a$
 $30 = 5 \cdot a$
 $a = \frac{30}{5} \Rightarrow \boxed{a = 6 \text{ m/s}^2}$

* $v = v_0 + at$ ($v_0 = 0$)
 $v = 6 \cdot 5$
 $\boxed{v = 30 \text{ m/s}}$



5. $m = 10^3 \text{ kg}$
 $v_0 = 72 \text{ km/h} = 20 \text{ m/s}$
 $\Delta s = 50 \text{ m}$



* $v^2 = v_0^2 + 2a\Delta s$
 $0 = 20^2 + 2 \cdot a \cdot 50$
 $100 \cdot a = -400$
 $a = \frac{-400}{100} \Rightarrow a = -4 \text{ m/s}^2$

a) $v = v_0 + at$
 $0 = 20 - 4 \cdot t$
 $4t = 20$
 $t = \frac{20}{4} \Rightarrow \boxed{t = 5 \text{ s}}$

b) $F_R = m \cdot a$
 $F_R = 10^3 \cdot 4 \Rightarrow \boxed{F_R = 4 \cdot 10^3 \text{ N}}$

6. $\Delta S = 400 \text{ m}$

$v_0 = 20 \text{ m/s}$

$v = 40 \text{ m/s}$

$m = 1200 \text{ kg}$

* $v^2 = v_0^2 + 2 \cdot a \cdot \Delta S$

$40^2 = 20^2 + 2 \cdot a \cdot 400$

$1600 = 400 + 800 \cdot a$

$800 \cdot a = 1200$

$a = \frac{1200}{800} \Rightarrow a = 1.5 \text{ m/s}^2$

* $F_p = m \cdot a$

$F_p = 1200 \cdot 1.5$

$F_p = 1800 \text{ N}$

7. $F = 2 \text{ N}$

$v_0 = 0$; $t = 10 \text{ s}$

$m = 500 \text{ g} = 0.5 \text{ kg}$

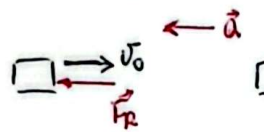
* $F_R = m \cdot a \Rightarrow a = 0.5 \cdot a \Rightarrow a = \frac{2}{0.5} \Rightarrow a = 4 \text{ m/s}^2$

* $\Delta S = v_0 \cdot t + \frac{a}{2} t^2 \Rightarrow \Delta S = \frac{4}{2} \cdot 10^2 \Rightarrow \Delta S = 200 \text{ m}$

8. $t = 4 \text{ s}$

$m = 8 \text{ kg}$

$v_0 = 9 \text{ m/s}$; $v = 6 \text{ m/s}$



* $v = v_0 + at \Rightarrow 6 = 9 + a \cdot 4 \Rightarrow 4a = 6 - 9 \Rightarrow$

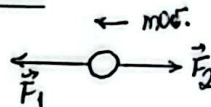
$\Rightarrow 4a = -3 \Rightarrow a = -\frac{3}{4} \Rightarrow a = -0.75 \text{ m/s}^2$

* $F_R = m \cdot a \Rightarrow F_R = 8 \cdot 0.75 \Rightarrow F_R = 6 \text{ N}$ (sentido oposto ao do movimento)

9. $m = 5 \text{ kg}$

$F_2 = 24 \text{ N}$

* MRU $\Rightarrow (F_R = 0 \text{ e } F_R = 0)$



* $F_1 - F_2 = 0 \Rightarrow F_1 = F_2$

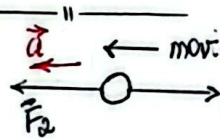
$F_1 = 24 \text{ N}$

10. $m = 4 \text{ kg}$

$F_1 = 70 \text{ N}$

$a = 5 \text{ m/s}^2$

MRUV acelerado



$\Rightarrow a$ para a esquerda $\therefore F_R$ é para a esquerda e $F_2 > F_1$

$F_R = m \cdot a$

$F_2 - F_1 = m \cdot a \Rightarrow F_2 - 70 = 4 \cdot 5 \Rightarrow F_2 - 70 = 20$

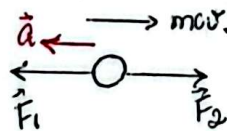
$F_2 = 70 + 20 \Rightarrow F_2 = 90 \text{ N}$

11. $m = 6 \text{ kg}$

$a = 3 \text{ m/s}^2$

$F_2 = 50 \text{ N}$

MRUV retardado



$\Rightarrow a$ para a esquerda $\therefore F_R$ é para a esquerda $\therefore F_1 > F_2$

$F_R = m \cdot a \Rightarrow F_1 - F_2 = m \cdot a \Rightarrow F_1 - 50 = 6 \cdot 3$

$F_1 = 18 + 50 \Rightarrow F_1 = 68 \text{ N}$